Ramakrishnan Sundara Raman

Assistant Professor, Computer Science and Engineering, University of California Santa Cruz

rsundar2@ucsc.edu Personal Website LinkedIn Google Scholar Lab Website

Positions

- **University of California Santa Cruz**
  Department of Computer Science and Engineering, Baskin School of Engineering
  Assistant Professor Jul 2024 – Present
- **Cloudflare**, *Research Intern* Jul 2022 – Sep 2022
- **The Citizen Lab**, *OTF Senior ICFP Fellow* Nov 2021 – Jul 2022
- **Microsoft**, *Associate Consultant* Jul 2017 – May 2018
- **University of Maryland**, *Research Intern* Mar 2017 – Jul 2017

Education

2018 – 2024 **Ph.D. in Computer Science, University of Michigan** in Computer Science and Engineering
Advisor: Prof. Roya Ensafi

2013 – 2017 **Bachelor of Technology, Vellore Institute of Technology** in Computer Science and Engineering
Project Title: Measuring Last-Mile Internet Reliability During Severe Weather

Conference Publications

- **Evaluating Practical Enumeration and Blocking Attacks on the Snowflake Circumvention System**
  L. Chen, R. Sangha, C. Bocovich, **R. Sundara Raman**
  ACM SIGSAC Conference on Computer and Communications Security (CCS), November 2026
- **PrivAudit: A Dual-Lens Auditing Framework for Website Privacy Practices under the CCPA**
  M.M. Dawoud, R. Aggarwal, L.R. Krishnamoorthy, **R. Sundara Raman**
  ACM SIGSAC Conference on Computer and Communications Security (CCS), November 2026
- **CenRL: A Framework for Performing Intelligent Censorship Measurements**
  H. Le, A. Huremagic, K. Wang, R. Ensafi, **R. Sundara Raman**
  IEEE Security and Privacy (S&P), May 2026
- **Fingerprinting Deep Packet Inspection Devices by their Ambiguities**
  D. Xue, A. Huremagic, W. Wang, **R. Sundara Raman**, R. Ensafi
  ACM SIGSAC Conference on Computer and Communications Security (CCS), October 2025
  IRTF Applied Networking Research Prize
- **Digital Discrimination of Users in Sanctioned States: The Case of the Cuba Embargo**
  A. Ablove, S. Chandrashekaran, H. Le, **R. Sundara Raman**, R. Ramesh, H. Oppenheimer, R. Ensafi
  USENIX Security Symposium, August 2024
  Distinguished Paper Award FOCI Best Practical Award
- **Modeling and Detecting Internet Censorship Events**
  E. Tsai, **R. Sundara Raman**, A. Prakash, R. Ensafi
  Network and Distributed System Security Symposium (NDSS), February 2024
- **Global, Passive Detection of Connection Tampering**
  R. Sundara Raman, L.H. Merino, K. Bock, M. Fayed, D. Levin, N. Sullivan, L. Valenta
  ACM SIGCOMM, September 2023
- **Network Responses to Russia's Invasion of Ukraine in 2022: A Cautionary Tale for Internet Freedom**
  R. Ramesh*, **R. Sundara Raman***, A. Virkud, A. Dirksen, A. Huremagic, D. Fifield, D. Rodenburg, R. Hynes, D. Madory, R. Ensafi
  USENIX Security Symposium, August 2023

*Both authors contributed equally to this research

FOCI Best Practical Award 2024

- **CERTainty: Detecting DNS Manipulation using TLS Certificates**
  E. Tsai, D. Kumar, **R. Sundara Raman**, G. Li, Y. Eiger, R. Ensafi
  Privacy Enhancing Technologies Symposium (PETS), July 2023

- **Network Measurement Methods for Locating and Examining Censorship Devices**
  R. Sundara Raman*, M. Wang*, J.Dalek, J. Mayer, and R. Ensafi
  ACM Conference on emerging Networking EXperiments and Technologies (CoNEXT), December 2022
  IRTF Applied Networking Research Prize

- **A Large-scale Investigation into Geodifferences in Mobile Apps**
  R. Kumar, A. Virkud, **R. Sundara Raman**, A. Prakash, and R. Ensafi
  USENIX Security Symposium, August 2022

- **Lost in Transmission: Investigating Filtering of COVID-19 Websites**
  A. Vyas, **R. Sundara Raman**, N. Ceccio, P. Lutscher, and R. Ensafi
  Financial Cryptography and Data Security (FC), March 2021

- **Censored Planet: An Internet-wide, Longitudinal Censorship Observatory**
  R. Sundara Raman, P. Shenoy, K. Kohls, and R. Ensafi
  ACM SIGSAC Conference on Computer and Communications Security (CCS), November 2020

- **Investigating Large Scale HTTPS Interception in Kazakhstan**
  R. Sundara Raman, L. Evdokimov, E. Wustrow, A. Halderman, and R. Ensafi
  ACM Internet Measurement Conference (IMC), October 2020
  Best Paper Award Nominee

- **Measuring the Deployment of Network Censorship Filters at Global Scale**
  R. Sundara Raman, A. Stoll, J. Dalek, R. Ramesh, W. Scott, R. Ensafi
  Network and Distributed System Security Symposium (NDSS), February 2020

- **Decentralized Control: A Case Study of Russia**
  R. Ramesh, **R. Sundara Raman**, M. Bernhard, V. Ongkowijaya, L. Evdokimov, A. Edmundson, S. Sprecher, M. Ikram, R. Ensafi
  Network and Distributed System Security Symposium (NDSS), February 2020
  CSAW'20 Applied Research Competition North America Top 10 Paper

Workshop Publications

- **From Underground to Mainstream Marketplaces: Measuring AI-Enabled NSFW Deepfakes on Fiverr**
  M. M. Dawoud, A. Cuevas, **R. Sundara Raman**
  Symposium on Usable Security and Privacy (USEC), February 2026

- **Not What It Used To Be: Generational Analysis of Top-level Domain Reputation**
  J. Szurdi, R. Ramesh, **R. Sundara Raman**, D. Liu
  Workshop on Measurements, Attacks, and Defenses for the Web (MADWeb) 2026
  Distinguished Paper Award

- **Advancing the Art of Censorship Data Analysis**
  R. Sundara Raman, A. Virkud, S. Laplante, V. Fortuna, R. Ensafi
  Free and Open Communications on the Internet (FOCI), February 2023

- **A Multi-Perspective View of Internet Censorship in Myanmar**
  R. Padmanabhan, A. Filastò, M. Xynou, **R. Sundara Raman**, K. Middleton, M. Zhang, D. Madory, M. Roberts, and A. Dainotti
  Free and Open Communications on the Internet (FOCI), August 2021

Awards

Research — **MADWeb 2026 Best Paper Award**
February 2026

- **IRTF Applied Networking Research Prize (ANRP) 2026**
  January 2026
- **USENIX Security 2024 Distinguished Paper Award Award**
  August 2024
- **FOCI'24 Best Practical Award**
  July 2024
- **FOCI'23 Rising Star Award**
  July 2023
- **IRTF Applied Networking Research Prize (ANRP) 2023**
  January 2023
- **University of Michigan CSE Graduate Student Honors Runners Up Award**
  November 2020
- **ACM IMC 2020 Best Paper Award Nominee**
  October 2020

Fellowships

- **Rackham Predoctoral Fellowship**
  March 2023
- **OTF Information Controls Fellowship (ICFP)**
  November 2021

Teaching & Service

- **North Campus Deans' MLK Spirit Award 2023**
  January 2023
- **University of Michigan CSE GSI Honorable Mention Award**
  August 2021

Student Awards

- **Rohit Nagubandi, Tor SEEKCommons Fellowship**
  February 2025
- **Sriya Katreddi, Deans Award for Undergraduate Research**
  June 2025
- **Prasiddh Pooskur, Deans Award for Undergraduate Research**
  June 2025

Mentoring

- **PhD Students:** Linden Chen (2025-Present), Mohamed Moustafa Dawoud (2025-Present), Ramon Torres (2025-Present)
- **PostDocs/Visiting Researchers:** Kevin Wang (2026-Present)
- **Masters Students:** Pakhi Sinha (2025-2026), Kevin Tai (2026), Rohit Nagubandi (2024-2026), Riya Aggarwal (2025), Likith Rahul Krishnamurthy (2025), Kevin Wang (Michigan, 2023)
- **Undergraduate Students:** Nate Jarrett (2026-Present), Harshita Sarathy (2025-Present), Alexus Huang (2026-Present), Rachel An (2026-Present), Sofia Yakushi (2026-Present), Yohann Moraes (2026-Present), Sonia Sarkar (2026-Present), Siddhant Madhur (2026-Present), Pavitra Vivekanandan (2026-Present), Kelvin Chan (2026-Present), Anvie Swaroop (2026-Present), Isa Abello (2025-2026), Andrew Chu (2025-2026), Geoffrey Tang (2025-2026), Ethan Cao (2026), Graham Klinger (2025-2026), Mukesh Saravanan (2025-2026), Kenneth Lai (2025-2026), Michael Kamensky (2025-2026), Sriya Katreddi (2024-2026), Ruthwika Gajjala (2026), Shubham Marathe (2026), Rachit Verma (2026), Aaryan Ahlawat (2025), Udbhav Sikarwar (IIT Kanpur, 2025), Peter Dobbins (2025), Felicity Shih (2024-2025), Carlo Aqui (2025), Shivani Belambe (2025), Saurav Shah (2025), Dominick Rangel (2024-2025), Leonardo Gallego (2024), Brian Huang (2024-2025), Prasiddh Pooskur (2024-2025), Ryan Sangha (2024-2025), Nadav Oren (Michigan, 2023), Apurva Virkud (Michigan, 2020-2022), Anjali Vyas (Michigan, 2020-2021), Prerana Shenoy (Michigan, 2019-2020)
- **Lamat Institute:** Daisy Vanesa Nabor Escamilla (2026)
- **Cal-Bridge:** John Alora (2025-Present), Noah Lee (2026-Present)

Teaching

Winter 2025, 2026	**Instructor, CSE 153, UC Santa Cruz** Topic: Network Security and Privacy (Undergraduate)
Fall 2024, 2025, 2026	**Instructor, CSE 253, UC Santa Cruz** Topic: Network Security (Graduate)
Spring 2025	**Instructor, CSE 156, UC Santa Cruz** Topic: Network Programming (Undergraduate)
March 2023	**Guest Lecturer, CS 4973, Northeastern University** Topic: Censorship and Circumvention
January 2023	**Substitute Instructor, EECS 588, University of Michigan** Topic: Computer and Network Security
March 2022	**Guest Lecturer, EECS 588, University of Michigan** Topic: Censorship and Circumvention
Fall 2020	**Graduate Student Instructor (GSI), EECS 588, University of Michigan** Topic: Computer and Network Security GSI Honorable Mention Award

Talks

- **Invited Panel: Tech&Democracy**
  Encuentro Digital, The Hispanic Federation, September 2026
- **Conference Talk: CenRL: A Framework for Performing Intelligent Censorship Measurements**
  IEEE S&P, May 2026
- **Invited Talk: Advancing the Art of Internet Censorship Measurement**
  Dartmouth College, October 2025
- **Invited Talk: Network Measurement Methods for Locating and Examining Network Censorship Devices**
  IETF 118, November 2023
- **Conference Talk: Global, Passive Detection of Connection Tampering**
  ACM SIGCOMM, September 2023
- **Conference Talk: Network Responses to Russia's Invasion of Ukraine in 2022: A Cautionary Tale for Internet Freedom**
  Usenix Security Symposium, August 2023
- **Workshop Talk: Advancing the Art of Censorship Data Analysis**
  Free and Open Communications on the Internet (FOCI), February 2023
- **Lightning Talk: Exploring Internet censorship through the Censored Planet Dashboard**
  Open Technology Fund (OTF), January 2023
- **Conference Talk: Network Measurement Methods for Locating and Examining Network Censorship Devices**
  Conference on emerging Networking EXperiments and Technologies (CoNEXT 2022), December 2022
- **Tech Demo: Exploring Internet censorship with the Censored Planet Dashboard**
  RightsCon 2022, June 2022
- **Symposium Talk: Censored Planet: An Internet-wide, Longitudinal Censorship Observatory**
  The 5th Annual Data for Public Good Symposium, March 2022
- **Invited Talk: Censored Planet Community Webinar 2021**
  Censored Planet, October 2021
- **Invited Talk: Censored Planet: An Internet-wide Longitudinal Observatory**
  NDSS DNS Privacy Workshop (DNSPriv), February 2021
- **Conference Talk: Censored Planet: An Internet-wide Longitudinal Observatory**
  ACM Conference on Computer and Communications Security (CCS), November 2020

- **Conference Talk: Investigating Large Scale HTTPS Interception in Kazakhstan**
  ACM Internet Measurement Conference (IMC), October 2020
- **Invited Talk: Censored Planet: Measuring Internet censorship remotely**
  OONI Internet Measurement Village (IMV), June 2020
- **Conference Talk: Measuring the Deployment of Network Censorship Filters at Global Scale**
  Network and Distributed System Security Symposium (NDSS), February 2020

Reports and Blogposts

- **A Swift Look into the Internet Allowlist in Iran**
  RANDLab Blog, June 2026
  https://randlab.engineering.ucsc.edu/blogs/iran-allowlist/
- **Insights from data at large scale might just be a small miracle**
  The Cloudflare Blog, October 2025
  https://blog.cloudflare.com/experience-of-data-at-scale/#insights-from-data-at-large-scale-might-just-be-a-small-miracle
- **A Global Assessment of Third-Party Connection Tampering**
  The Cloudflare Blog, September 2024
  https://blog.cloudflare.com/connection-tampering/
- **Measuring Connection Tampering Around the World**
  Internet Society Pulse Blog, October 2023
  https://pulse.internetsociety.org/blog/measuring-connection-tampering-around-the-world
- **Making Censorship Data More User Friendly**
  Internet Society Pulse Blog, May 2023
  https://pulse.internetsociety.org/blog/making-censorship-data-more-user-friendly
- **Profiling Internet Censors: Methods to Measure and Locate Censorship Devices**
  Open Technology Fund (OTF), December 2022
  https://www.opentech.fund/news/profiling-internet-censors-methods-to-measure-and-locate-censorship-devices/
- **An analysis of a large-scale HTTPS interception**
  APNIC Blog, February 2021
  https://blog.apnic.net/2021/02/15/an-analysis-of-a-large-scale-https-interception/
- **US Government and military websites are geoblocked from Hong Kong and China**
  Censored Planet, August 2020
  https://censoredplanet.org/hongkong

Service

Chairing
- **Program Co-Chair at Free and Open Communications on the Internet (FOCI) 2027-2028**
- **Session Chair at ACM Internet Measurement Conference (IMC) 2025**
- **Session Chair at Free and Open Communications on the Internet (FOCI) 2021**

Reviewing
- **TPC Member, ACM SICOMM Education Workshop 2026**
- **TPC Member, ACM Conference on Computer and Communications Security (CCS) 2026**
- **TPC Member, ACM Internet Measurement Conference (IMC) 2026**
- **Reviewer, IEEE/ACM Transactions on Networking (TON) 2025**
- **TPC Member, ACM Internet Measurement Conference (IMC) 2025**
- **TPC Member, Free and Open Communications on the Internet (FOCI) 2025**
- **TPC Member, IEEE Symposium on Security and Privacy (S&P) 2025**
- **TPC Member, ACM Internet Measurement Conference (IMC) 2024**
- **TPC Member, IEEE Symposium on Security and Privacy (S&P) 2024**
- **TPC Member, Free and Open Communications on the Internet (FOCI) 2024**

- **Reviewer, 6th International Conference on Computer Science and Application Engineering (CSAE) 2022**
- **TPC Member, ACM SIGCOMM 2021 Workshop on Free and Open Communications on the Internet (FOCI)**

Leadership

- **Member of Committee on Information Technology (CIT), UCSC**
  September 2026 – Present
- **Security Seminar Host, UCSC**
  September 2024 – Present
- **Outreach Chair at CSEG, UM CSE graduate student organization**
  June 2020 – May 2022
  My responsibilities included arranging and managing outreach events throughout the year that aimed at spreading knowledge about computer science research in various schools, colleges and groups.
  Deans' MLK Spirit Award 2023